

S1.1: Discrete Random Variables

1. Probability Distributions

A **discrete random variable** X takes a countable set of values $x_1, x_2, \dots$. Its probability mass function (pmf) is $P(X = x_i) = p_i$, with

$$\sum_i p_i = 1, \quad 0 \leq p_i \leq 1.$$

X has pmf given by $P(X = x) = kx$ for $x = 1, 2, 3, 4$. Find k .
 $k(1 + 2 + 3 + 4) = 1 \Rightarrow 10k = 1 \Rightarrow k = 0.1$.

2. Expectation and Variance

$$E(X) = \mu = \sum_i x_i p_i$$

$$\text{Var}(X) = \sigma^2 = E(X^2) - \mu^2 = \sum_i x_i^2 p_i - \mu^2$$

$$E[g(X)] = \sum_i g(x_i) p_i$$

Linear Transformations

$$E(aX + b) = aE(X) + b$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Note: a constant shift doesn't affect variance; a scaling factor is squared.

3. Sums of Independent Discrete RVs

For *independent* X and Y :

$$E(X + Y) = E(X) + E(Y)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$ — *add* variances even when subtracting (for independent variables). A very common slip.

General Linear Combination

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) \quad (\text{independent})$$

Sum of n IID Copies vs. Scaling

If $X_1, \dots, X_n$ are IID copies of X , and $S = X_1 + \dots + X_n$:

- $E(S) = nE(X)$, $\text{Var}(S) = n \text{Var}(X)$.

Contrast: $Y = nX$ (a single X scaled by n):

- $E(Y) = nE(X)$, $\text{Var}(Y) = n^2 \text{Var}(X)$.

OCR Exam Tip: Distinguish between "the total from n rolls" (sum) and " $10 \times$ (single roll)" (scaling). Totals from independent trials have n variance; scaling has n^2 .

4. The Discrete Uniform Distribution

$X \sim U\{1, 2, \dots, n\}$: each outcome equally likely with probability $1/n$.

- $E(X) = \frac{n+1}{2}$
- $\text{Var}(X) = \frac{n^2-1}{12}$

5. Joint Discrete Distributions

Two discrete random variables X, Y have a **joint pmf** $P(X = x, Y = y)$, often displayed as a table.

Marginals

$$P(X = x) = \sum_y P(X = x, Y = y), \quad P(Y = y) = \sum_x P(X = x, Y = y)$$

Independence

$$X \perp Y \iff P(X = x, Y = y) = P(X = x)P(Y = y) \text{ for all } x, y.$$

Conditional Distribution

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

Joint table for (X, Y) :

$X \backslash Y$	0	1	2
0	0.1	0.2	0.1
1	0.2	0.3	0.1

Marginals: $P(X = 0) = 0.4$, $P(X = 1) = 0.6$. $P(Y = 0) = 0.3$, $P(Y = 1) = 0.5$, $P(Y = 2) = 0.2$.

Independence? Check $P(X = 0, Y = 0) = 0.1$ vs. $P(X = 0)P(Y = 0) = 0.12$. Not equal — so X, Y are **not independent**.

6. Covariance (Support)

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

For general (not necessarily independent) X, Y :

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$$

$X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$ (the converse is not generally true).

7. Exam Style Strategy

- Use the formula $\text{Var}(X) = E(X^2) - \mu^2$ — computing $E[(X - \mu)^2]$ directly is error-prone.
- For distributions with a parameter k , use $\sum p_i = 1$ to find k first.
- Drawing a joint table is very helpful; always check marginals sum correctly.
- When working with $aX + bY$, track signs of a, b carefully; in variance they are squared and so always non-negative.